

SARVARBEK BOTIROV

sarvarb808@gmail.com
Seoul, South Korea

PROFILE SUMMARY

Master's student in Industrial Engineering at Yonsei University with strong expertise in procurement analytics, supply chain risk assessment, and data-driven decision making. Experienced in supplier coordination, cross-border communication, and cost analysis through hands-on work with international manufacturers. Skilled in Python, Gurobi, and Excel for optimization and market analysis. Focused on applying supply chain optimization to semiconductor sourcing challenges including geopolitical risk and supplier concentration.

EDUCATION

Master's degree | Industrial Engineering March 2024 – Aug 2026 (Expected)
Yonsei University Sinchon, Republic of Korea
GPA | 4.05/4.3

- Research focus: supply chain analytics & optimization, green investment decisions, carbon emission policies, game-theoretic modeling under demand uncertainty
- Research Assistant, Smart Manufacturing & Operations Management (SMOM) Lab

Bachelor of Logistics Sept 2019 – Aug 2023
Inha University Incheon, Republic of Korea
GPA | 3.5/4.5

- Asia Pacific School of Logistics — coursework in global SCM, freight, and international trade operations

WORK EXPERIENCE

Research Assistant March 2024 – Present
SMOM Lab of Yonsei

- Developed stochastic demand models to analyze supply chain variability, identifying key factors affecting inventory and pricing decisions
- Conducted supply chain risk analysis under demand and policy changes, supporting strategic planning insights
- Applied optimization tools (Python, Gurobi) to solve cost and efficiency problems in supply chain systems
- Collaborated on research related to carbon regulation, green investment, and dual-channel supply chain strategies

Admissions Assistant Sep 2022 – Jan 2023
Admission Team, Inha University – Part time South Korea

- Managed Excel-based admissions database for 200+ applicants, ensured data accuracy using validation and vlookup functions
- Delivered trilingual document translation (English / Korean / Uzbek) ensuring clear cross-cultural communication

Sales & Supplier Coordinator Jan 2016 – Feb 2020
Family-Owned Textile Company Tashkent, Uzbekistan

- Coordinated with Chinese suppliers on product specifications, pricing, and delivery schedules to ensure order accuracy and timely fulfillment
- Assisted in negotiating delivery terms and quality requirements, improving clarity and reducing potential supply risks
- Supported procurement activities by managing supplier communication and resolving cross-border operational issues

PROJECTS AND RESEARCH

Supply Chain Analytics & Optimization (Thesis Research) Jan 2026 –
Yonsei University (SMOM Lab)

- Modeling supply chain strategy integrating green investment decisions, carbon emission policies, and game-theoretic optimization under demand uncertainty

Advertisement-Driven Demand in Stochastic Barter-Exchange Model for Retail Supply Chains Jan 2025 – Nov 2025

Yonsei University (SMOM Lab)

- Modeled retail supply chain demand variability and generated actionable insights using stochastic optimization frameworks (Python, Gurobi)

SmartForm: Intelligent Dumbbell Posture-Monitoring System Oct 2024 – Dec 2024

Yonsei University (Smart Manufacturing class)

- Course project, Fall 2024: developed AI-based sensor system with real-time data analytics pipeline using Python and ML frameworks

INDUSTRY EXPOSURE & TRAINING

K-Export Academy — Trade Specialist Track Aug 2025 – Nov 2025

Pusan University of Foreign Studies

- Completed 200-hour program covering international trade fundamentals, export policy analysis, global market intelligence, and supply chain risk assessment

Samsung C&T Discover Program July 2024

Samsung C&T (삼성물산)

- Selected for five-day corporate discovery program; visited NRD-K high-tech manufacturing and testing facility, Samsung Safety Academy, Samsung Electronics headquarters, and Samsung FAB Clean Room
- Attended sessions at Samsung University on corporate history, core values, and strategic vision; gained direct insight into procurement operations and quality standards in semiconductor manufacturing

LG Science Park Discovery Tour Aug 2025

LG

- Selected among international STEM graduate students for LG's innovation and smart manufacturing discovery program

CERTIFICATIONS

K-Export Academy — Trade Specialist Track (200 hrs)	Pusan University of Foreign Studies	Nov 2025
Machine Learning	DeepLearning.AI	Feb 2025
AWS Certified Cloud Practitioner	Amazon Web Services (valid Aug 2026)	Aug 2023
Data Engineer Track	DataCamp	Sep 2023
Data Analyst in SQL Track	DataCamp	Mar 2023

AWARDS & SCHOLARSHIPS

연세대 외국인특별장학금 (Yonsei Special International Scholarship)	Feb 2025
BK21 Brain Korea 21 Research Scholarship — Smart Manufacturing	Sep 2025 – Feb 2026
국제공동연구장학금 (International Joint Research Scholarship), Yonsei	Sep 2024
혁신재학조교장학금 GSRA (Graduate Student Research Assistantship)	2024 – Present
Scholarship for Outstanding International Students (I), Yonsei University	Mar 2024
Full Scholarship for Academic Excellence, Inha University	Jun 2022
Academic Excellence Scholarship, Inha University	Aug 2020

SKILLS

Technical skills:

Python (NumPy, Pandas, Matplotlib, sklearn), SQL
Gurobi (Mathematical Optimization), Mathematica

Microsoft Excel (Advanced functions, Data analysis)

Languages:

Korean (Topik 6), English (TOEIC 930), Russian (Intermediate), Uzbek (Native)

LEADERSHIP & VOLUNTEERING

Chairman, International Students Union

Aug 2022 – Feb 2023

Inha University

- Led the international student representative body; organized events and served as the main point of contact between international students and university administration

Buddy Program Mentor

Aug 2022 – Mar 2023

Inha University

- Mentored one incoming international exchange student for a full semester; supported academic orientation and cultural adjustment